Fairness in fNIRS: How Underrepresentation Affects Augmentation

Abstract

Generative Adversarial Networks (GANs) have shown promise in generating synthetic data to augment small datasets, such as time-series data for functional near-infrared spectroscopy (fNIRS). While artificial data can enhance classification accuracy, enable collaboration on projects involving sensitive data, and improve data visualizations, there is concern about amplification of existing biases, potential quality-of-service harms, and the ramifications for healthcare equity. In this project, underrepresented groups in fNIRS research will be identified through unstructured, focused interviews, along with suggestions for overcoming challenges in diverse participant recruitment. To demonstrate the impact underrepresentation may have on machine learning applications, a dataset using left-hand-tapping, right-hand-tapping, and foot-tapping will be used to generate synthetic data for classification and prediction. The hypothesis is that any bias in this dataset may be generalized to other demographics, such as those with curly hair or dark skin. The goals are to learn which demographics are underrepresented in neuroimaging research, to explore whether a time-series generator is biased towards traits that occur more frequently in the training data, and to learn about how fairness can propagate to downstream applications.

Introduction and Background

Fairness in fNIRS Research

Functional near-infrared spectroscopy (fNIRS) is a portable, non-invasive method of measuring the brain's changes in hemoglobin concentrations using infrared light. Applications include diagnoses of cognitive disorders, understanding of disease progression, and assisting those with disabilities [8]. However, there is a systemic underreporting of demographics in literature, though factors such as hair type, melanin levels, hair length, and more have been known predictors of data quality for upwards of 25 years [11][12]. Of 69 fNIRS studies that report reasons for participant exclusion, 11.6% cite hair as a reason [11], and especially thick or dark hair. Even when participants with difficult hair types are measured, their results are often discarded due to poor data quality [19]. This methodological, phenotypic bias can contribute to medical mistrust and inability to test for fairness in these studies, since demographic information is not widely reported. By identifying practical challenges researchers face in diverse participant recruitment, we may better understand how to improve best practices in the field.

Machine Learning in fNIRS Research

Machine learning (ML) is used widely in fNIRS research for data preprocessing, analysis, modeling, and more. As of 2022, eight papers used deep learning (DL) for preprocessing or augmentation, eleven papers used DL for cortical analysis, fourteen used DL for diagnosis, and thirty used DL for task identification based on cortical activation (brain-computer interfacing) [2]. Each of these steps introduce fairness challenges that researchers should consider. Preprocessing is a step that removes physiological signals, such as heartbeat or breathing, from the reading to isolate the oxygenated hemoglobin levels. At present, there are inadequate open-source tools, documentation, and standards, resulting in potential false results and difficulty in verification [19]. For example, the pathlength factor parameter (used to model the hemoglobin concentration) must be modified based on age of participant [19]; this can cause the results for older adults and children to be modelled unusually poorly. DL can also be used for automated feature extraction; one study reported that a multi-layer perceptron achieved a classification accuracy of 78% to 80% on four classes using no denoising or preprocessing steps [2]. While this demonstrates the effectiveness of an ML approach, model transparency is lost. This affects the medical trust of marginalized populations in particular due to historic harm; any clinical applications should ensure their results are explainable and fair to all patients.

In addition to the aforementioned challenges, fNIRS analysis is also limited by small sample size – this can compound the difficulty of recruiting diverse participants. Since human subjects research is time-consuming and expensive, researchers have recently begun to use tools such as generative adversarial networks (GANs) to generate synthetic time-series data in a process known as data augmentation. Artificially increasing the size of the dataset has been shown to improve classification accuracy in fNIRS data [2][6] and the quality of explainable visualizations [6]. Additionally, the synthetic data can serve as a proxy dataset, enabling collaboration while limiting privacy concerns [5][6]. However, these benefits are only available to individuals who are accurately represented in the resulting synthetic data; the goal of this project is to identify which groups are underrepresented in fNIRS research, then to explore how this imbalance in training data may affect the quality and utility of synthetic data.

Fairness in GANs

Fairness research, especially as applied to generative AI and neuroimaging methodologies, is a relatively new area of study, but one critical to automated image recognition and prevention of quality-of-service harm. Bias in deep learning models can produce statistically-significant performance gaps between gender, ethic, and age subgroups, leading to concerns about health care equity and a model's potential to amplify existing bias [5][7].

To mitigate these concerns, it is important to develop generative models that represent real datasets while limiting bias. Generated data should meet the following requirements: (1) data utility, (2) data fairness, (3) classification utility, and (4) classification fairness [3]. Some models, such as FairGAN, have been developed to meet these requirements and achieve statistical parity in synthetically-generated data, but have not yet been generalized to produce time-series data. Ensuring fairness in the synthetic dataset improves fairness in downstream applications as well, such as predictive analysis or classification, so it is crucial that GANs can accurately represent data from different demographics of people.

Objectives and Research Questions

Project Objectives

This project has two goals: (1) investigate common practical challenges in data collection, then (2) explore how under-representation of various demographics might impact the quality of synthetic data meant to represent these individuals. To do this, I will meet the following objectives:

1. Summarize the practical challenges in data collection and analysis via user research to learn about which demographics may be under-represented in fNIRS studies

2. Demonstrate that under-representation of a demographic results in poorer quality of synthetic data for that demographic (hypothesis)
    a. Generate a time-series dataset from a limited, real-world dataset (using the normal distribution)
    b. Generate a time-series dataset from the same dataset, with one demographic significantly reduced
3. Learn about how fairness affects both synthetic data generation and downstream applications (such as classification or predictive analysis)
    a. Train a classifier for the normal distribution and reduced-demographic distribution, then compare the results

Research Questions

1. Which demographics are underrepresented in fNIRS studies?
2. What practical challenges do fNIRS researchers experience in data collection, preprocessing, and analysis? What can researchers do to make sure participation in these studies are accessible to everyone?
3. How does underrepresentation of a demographic in data collection affect the results of data augmentation? Will providing a model with less data from a certain demographic (compared to others) cause it to produce less representative synthetic data?

Methodology

User Research

To discover which practical challenges are most common in data collection and analysis, I am performing unstructured, focused interviews with fNIRS researchers in the state of Colorado. This type of interview does not have a set of predefined questions, and instead is guided by the researcher toward a relevant topic of interest [10]. This allowed me greater flexibility than structured interviews and more in-depth exploration. Additionally, the open-ended nature helped me reduce the bias that predefined questions may have introduced. Since the conversations were largely defined by the participants, my own perspective was not embedded into question-writing; this would have limited the types of viewpoints and interests the participants shared with me.

In total, I interviewed seven individuals. Three are graduate students working with the Imagine AI Lab [9] at the University of Colorado Boulder who have experience in data collection, analysis, and often other types of neuroimaging methods. Two participants were top researchers attending the fNIRS Workshop at CSU Spur – they provided insight into the current research landscape including trends, previous work, and recent breakthroughs. Two interviewees were previous participants in fNIRS pilot experiments, meaning they were familiar with the basic setup and experience of wearing the cap. Demographics are provided in Table 1. Of these seven interviews, four were conducted face-to-face, and three were conducted virtually by Zoom. Follow-up questions were sent by text or email, if necessary.

The interviews ranged from 15 to 30 minutes in length and were not always continuous. For example, the researchers at the fNIRS Workshop interviewed between research talks and over lunch break, but the total time spent discussing fairness-related topics fell into the reported range.

Topics covered in the interviews included previous projects, ways that studies could be made more accessible, challenges in everyday work, challenges in data collection, challenges in data preprocessing/analysis, how various factors would impact an individual's likelihood of participating in studies, and more. Detailed interview notes are provided in Appendix A. At the beginning or end of the interview, I described the nature of INFO 5871 and the associated term project to ensure each interviewee was effectively informed about the purpose of the study.

Synthetic Data – Model

Qualitative interviews can help identify underrepresented groups in fNIRS studies during data collection; however, it is still unclear what impact underrepresentation has on machine learning applications. To answer this question, I generate synthetic fNIRS data based on data from real participants (as described in the next section) to quantify the importance of representative training data with a generative adversarial network (GAN).

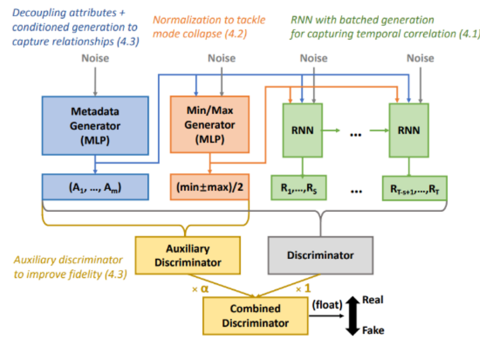

Figure 1: DoppelGANger architecture diagram

DoppelGANger [16] is a GAN that produces multivariate time-series data with corresponding metadata (such as channel and label), and is meant to accurately learn the temporal dependencies inherent in the dataset. This architecture takes less time to train than many comparable models (such as RCGAN, TimeGAN, etc.), but is comparable to (or better) in predictive accuracy and mean-squared error.

I use ydata-synthetic's package [17] in Google Colab to implement this architecture. Examples of the model may be found in the package's documentation: https://docs.synthetic.ydata.ai/1.2/examples/doppelganger_example/

Synthetic Data - Dataset

The open-access fNIRS dataset accessible on Github [14] consists of 30 voluntary participants. Each participant's cerebral hemoglobin patterns were tracked and quantified over the course of 75 trials. Of those 75 trials, 25 involved the participant taking part in left-hand unilateral complex tapping, 25 of them involved the participant taking part in right-hand unilateral complex tapping, and 25 of them involved the participant tapping their foot. Each of these trials involves the collection of data through 20 channels: 10 channels on the left cerebral hemisphere and 10 on the right. Each channel reports concentration changes for both oxygenated and deoxygenated hemoglobin. After preprocessing the data (using bandpass filtering, segmentation, and baseline correction), the authors extract features that consist of three epochs: spans of time from 0 - 5 seconds, 5 - 10 seconds, and 10 - 15 seconds of the trial. The block average of each of these epochs are saved for each trial and channel, producing a [75 x 120] dimension feature matrix for each participant.

The feature vectors and ground truth are produced by the classification script provided by the authors. Then, feature vectors for individual participants are rearranged into [75, 40, 3] numpy arrays. When aggregated over all participants, we produce a [2250, 40, 3] numpy array (as each of our thirty participants contains 75 trials).

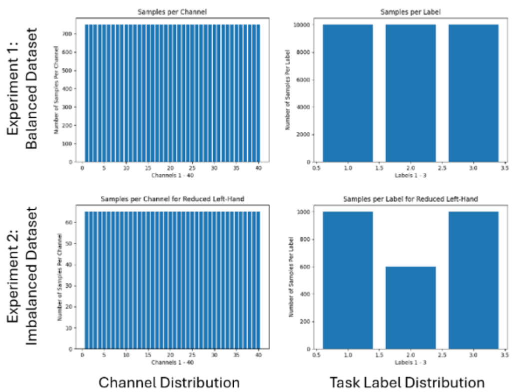

Figure 2: Distribution of synthetic data for Dataset 1 (top row) and Dataset 2 (bottom row). The distribution of channels is shown in the left column, and the distribution of the labels is shown on the right.

There are very few open-access fNIRS datasets that report demographic information, so in order to evaluate how bias propagates, I create an intentionally underrepresented group to compare to the original, balanced dataset (where the number of samples per task are equal). The balanced dataset will ensure the training dataset has an even distribution of each task: 33% left-finger-tapping, 33% right-finger-tapping, and 33% foot-tapping (Dataset 1). The latter will remove data from a task, producing a distribution of 38.46% right-finger-tapping, 38.46% foot-tapping, and 23.08% left-finger-tapping (Dataset 2). By training a different GAN model on each dataset, then comparing the results as described in the Evaluation and Testing subsection, I will be able to evaluate the effect of underrepresentation on the effectiveness of synthetic data for classification. See Figure 2 for demographic distribution.

Synthetic Data - Evaluation and Testing

The synthetic data is evaluated based on propagated bias, quality, and utility [18]. For the first two measures, I plot the demographic and label distributions (similarly to Figure 2). Data that follows the existing probability distribution indicates that bias in training data could propagate through data augmentation, unless care is taken to balance the demographics before or after the machine learning application. Missing channels or large variance may indicate poor data quality; even in real experiments, noisy and incomplete channels are usually removed before preprocessing and analysis of data. Evaluation of resulting demographic will help determine whether bias continues in augmented data, and whether bias in data collection can impact bias in data preprocessing and analysis.

To measure the data utility, I compare the classification and prediction accuracy of real data and synthetic data. Each task must contain 40 channels; therefore, I create training and validation datasets specifically for utility evaluation. The real dataset contains a balanced distribution of real participant data with 33% right-finger-tapping, 33% left-finger-tapping, and 33% foot-tapping. The synthetic data is a compilation of generated data from DoppelGANger with an equal number of channels per task; this dataset includes 35.79% right-finger-tapping, 31.7% left-finger-tapping, and 32.49% foot-tapping (Dataset 3), so is approximately balanced. Implementation is done with the time-series AI (tsai) package [4], which provides tools for univariate and multivariate analysis. The package loads InceptionTime [1] into a learner object that produces both classification and predictive analysis. Tsai's implementation of InceptionTime may be found here: https://github.com/timeseriesAI/tsai/blob/main/tsai/models/InceptionTime.py. My implementation of DoppelGANger, data preprocessing and formatting, plotting, and evaluation may be found here: https://github.com/anra8571/INFO5871FinalProject.

Results and Discussion

fNIRS researchers agreed with the literature that fairness work is severely lacking in this field – papers usually do not report demographic information, recruitment tends to be difficult, and fairness is often a secondary concern. Though there have been a few recent papers addressing this topic, best practices tend to focus on standardization of analysis rather than fairness. In order to better understand the challenges researchers face in data collection, a step that commonly introduces bias, I interviewed seven individuals and summarize their results below.

Practical Challenges – Data Collection

The single biggest predictor of data quality is hair type, which is expected for a data collection method that requires wearing a cap. Bald people are the ideal phenotype for this type of research; straight, thin, and wavy hair produce the next best results. Researchers agree that thick and curly hair types are the most difficult to work with, as ambient light can interfere with the reading when channels don't fully touch the skin of the scalp. However, this can create several fairness concerns – these participants might be turned away for their hair type altogether, and those that are included may have significant portions of data excluded due to quality issues. By excluding data from participants with thick and curly hair, certain ethnicities begin to dominate the dataset, which can lead to poorer performance on future applications for these individuals.

Don Rojas, a professor at Colorado State University (CSU), suggests cornrowing the hair – the montage can be arranged in lines matching the dimensions of the rows. Colby, a black individual with an afro, found this idea favorable – he reported that the researchers had clearly put effort into learning about how black hair works, and was pleased they had the forethought to use a professional. However, this solution can be expensive (both hiring the hairstylist and compensating the participant for the process) and would certainly require a longer research session to accommodate. While the cornrowing process may improve the data quality, Colby thought it may also provide another reason to exclude curly-haired people from the study. Realistically, gel is an easier solution. Using this method to smooth down hair has been employed with various success, but once again, burdens curly-haired people unfairly compared to their straight-haired counterparts.

Figure 3: Private room in Imagine AI Lab covered by paper for inclusive fNIRS

The data collection method also presents barriers to participation for religious women who do not show their hair in public. Mohsena, a Muslim woman, reported that more conservative women would be less likely to participate if they knew the study involved wearing a cap. Even open-minded women who are interested in participating would require several accommodations - data collection must be done by all female research staff in a private space, with no windows. In this way, accommodating individuals with varying religious backgrounds is more straightforward than those with varying hair types. The Imagine AI Lab has set up such a space (see Figure 3), but assumes the research team has (1) a separate room that can be dedicated to this research and (2) enough female researchers to schedule effectively with this population. Though this demographic is often not included in this field's fairness recommendations, researchers should make an attempt to advertise their accommodations processes in order to include these women.

Another challenge with text-based studies [15] is (1) the assumption of fluent English and (2) a cultural understanding of western trends and values. When a study referencing American media trends (such as children swallowing Tide pods) was replicated in Seoul, South Korea, participants had trouble rating their motivation to intervene; as stated by one participant, “our children don’t eat laundry detergent.” Individuals of different cultural backgrounds often have different levels of exposure to technology, different socioeconomic backgrounds, different beliefs about medical research, and other factors that can influence their willingness to participate. Not only do they likely perceive greater risk, but there may be less opportunity for those not representative of the cultural/ethnic majority [19]. Though most studies do not explicitly attempt to recruit a diverse set of participants, there are several accommodations that can help improve the participant experience for marginalized groups: better educational materials for fNIRS research (helping participants to be more informed), improved compensation structures (to make this research opportunity affordable), and intentional cross-cultural collaboration (to avoid injection of western-based biases in text-based, subjective studies).

Finally, the literature suggests that fNIRS is a more accessible mode of neuroimaging due to its mobility [19]. However, there are still some accessibility challenges associated with data collection – a participant must be able to sit at a desk for an extended period of time, read words on a screen, and use a mouse and keypad. One researcher, Leo, designed an experiment that investigated mixed-reality for driving. He tested visual cues, audio cues, and tactile cues for hazard perception; though his focus was primarily on the attention-shifting mechanism rather than the fairness aspect, this demonstrated successful integration of alternative sensory inputs. Data collection methods can also be improved to integrate with screen readers or to allow for verbal responses rather than keyboard input. Moving forward, improvements to software and increased awareness of these barriers may allow a greater number of disabled individuals to participate in future studies.

Despite these practical challenges, there are efforts to algorithmically reduce the noise in fNIRS readings, which may mean that more original data can be preserved for the understudied individuals who do participate. Though the primary intention is to maintain completeness of readings, improved fairness may be a secondary benefit. Additionally, some studies are making special efforts to recruit diverse participants. Nolan explains that this is to ensure the neural feedbacks and underpinnings are correctly captured for a variety of people; once again, fairness may be an additional benefit to the primary goal. I hope that by sharing these results, researchers publish work with more transparency, better understand the benefits of diversity in their datasets, and consider accommodations for underrepresented groups.

Overall, fairness seems to be a secondary concern for fNIRS researchers. Only one of the researchers I interviewed explicitly studied fairness, and the other six interviewees reported no fairness concerns in their current work. The lack of fairness considerations in participant recruitment can create unbalanced training datasets, meaning that results may not be generalizable to the population. Additionally, as I demonstrate in the next section, the imbalance may propagate through machine learning into other parts of the study, such as analysis, classification, and prediction.

Synthetic Data - Model Results

Now that the qualitative interviews have identified several underrepresented groups in fNIRS research, it is important to quantify the impact of bias in data collection. The generation of synthetic data will demonstrate how GANs propagate bias in the training set, how imbalance affects the resulting data quality, and how bias may affect the performance of downstream machine learning applications. I used the two datasets described in Methodology – the original, balanced dataset (Dataset 1), and the dataset with an intentionally unbalanced label distribution (Dataset 2) (see Figure 2). I then used DoppelGANger to generate an entirely new dataset based on each, and compared the resulting distribution (see Figure 4). For label data, DoppelGANger consistently follows the probability distribution of the training dataset – this suggests that any bias in the real-world dataset may propogate through data augmentation. Groups that are underrepresented in training data will continue to be underrepresented; even after machine learning is used to artifically increase the size of the dataset, the demographic distribution will remain constant. Additionally, the unbalanced dataset produced more missing channels (eight missing, as opposed to one missing channel when trained on the balanced dataset), meaning that the data was noisier. This suggests that number and distribution of samples in the training dataset may both be factors in resulting data quality.

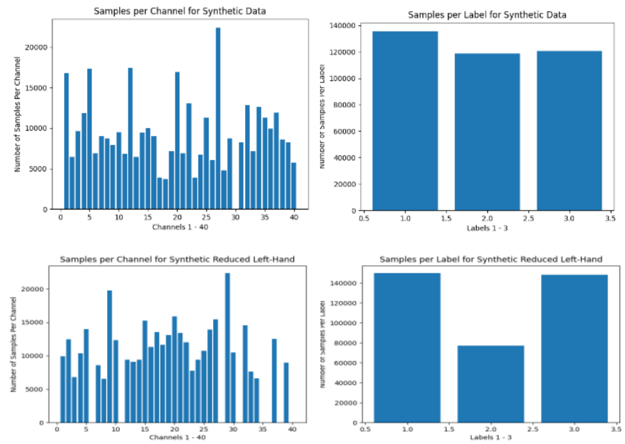

Figure 4: Distribution of synthetic data for balanced training data (top row) and balanced training data (bottom row). The distribution of channels is shown in the left column, and the distribution of the labels is shown on the right.

Another common method of evaluation is to use synthetic data for a downstream application, such as classification or prediction. This reveals how similar the synthetic data is to the real data, and how useful the synthetic data is on model training (often called data utility). To evaluate this, I generated enough data to create an additional, balanced synthetic dataset as described in

Synthetic Data - Evaluation and Testing to compare to a real dataset. InceptionTime produced poor classification results on both – the overall accuracy for the real dataset (Dataset 1) was 40%, and

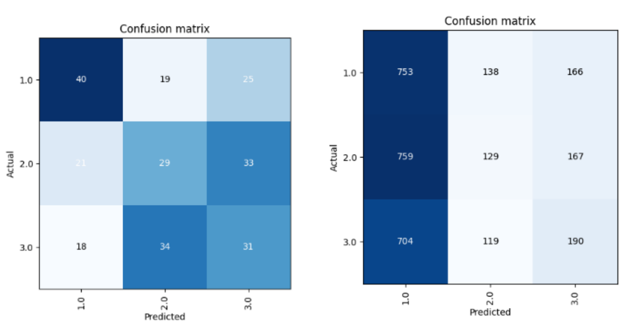

Figure 5: Confusion matrix for the real dataset (left) and synthetic dataset (right) on classification

	Class 1	Class 2	Class 3
F1 Score Real	0.4908	0.3515	0.36
F1 Score Synthetic	0.4601	0.179	0.247

Table 1: F1 Score for each class in the confusion matrices (see Figure 5)

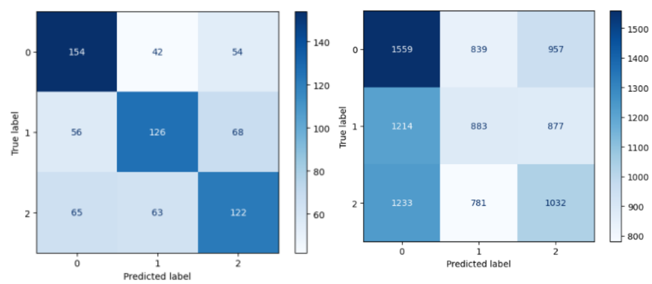

Figure 6: Confusion matrix for the real dataset (left) and synthetic dataset (right) on prediction

the synthetic dataset (Dataset 3) was 34.304% (see Figure 5). The F1 scores were also lower for every class in the synthetic dataset (see Table 1), indicating that the data quality decreased after augmentation for all task labels. InceptionTime was better at prediction. The predictive accuracy was 53.6% and 35.7%, for the real and synthetic datasets, respectively (see Figure 6 for the results per class).

The experiments in classification and prediction showed that the synthetic data utility was lower than real data – this means it is harder to use the synthetic data generated by DoppelGANger for other machine learning tasks. This is important because a common use case of GANs is to augment underrepresented demographics in training data; though this can increase accuracy by providing the model with an equal number of samples, this method may degrade the data quality if not carefully evaluated. Researchers should be careful when using synthetic data to avoid amplifying bias in demographic distribution, data quality of underrepresented demographics, and data utility.

Conclusion

Future Work

Given more time, a larger quantity of individuals could be interviewed for the qualitative research section. This would help me gather a more diverse set of perspectives, avoid generalization of a demographic based off limited information, and improve the credibility of results. A larger sample would be more convincing to the fNIRS community, helping to form more inclusive research recruitment and data analysis best practices.

I would also like to explore additional evaluation methods next semester, particularly in regards to statistical measurements and visual evaluation. These may provide additional insights regarding data utility, quality, and distribution that were difficult to discern in the reported evaluation methods.

Finally, I am interested in exploring synthetic data for augmentation of underrepresented groups. Rather than generating data for every participant, generating only underrepresented demographics could help to artificially balance the dataset. It would be interesting to create a third dataset by adding 400 data points to the left-hand-tapping task, then evaluating the newly augmented dataset using the reported methods.

Limitations and Challenges

The qualitative research was valuable, helping to identify fairness concerns in data collection that were not reported in the literature; however, there were limitations in participant recruitment. Sampling bias was the most notable – most participants were drawn from computer science backgrounds, which is not representative of fNIRS researchers as a whole (most have formal education in psychology and neuroscience). Additionally, the Imagine AI Lab has an unusually large population of international students. While this leads to improved cross-cultural research, it also means that the lab's graduate students may think about fairness more often than other fNIRS researchers, which introduces sampling bias.

Interviews were also conducted with only seven individuals; experiences may not be widely shared across demographics, so the results cannot be generalized to the broader population. In order to continue this type of human-subjects research, the lab would need to submit a proposal to the University's Internal Review Board (IRB), which can be a lengthy process. While it is ironic that the interviewees in this study are not representative of the total population, these preliminary results may inform more inclusive fNIRS best practices in the future.

There were also many challenges with TimeGAN, hence why DoppelGANger was used for the study instead. Notably, TimeGAN does not reconstruct full time-series sequences, but only generates windows within the original timeseries. These windows are not labelled; while this is okay for testing automated classification, it makes evaluation of fairness impossible. In both TimeGAN and DoppelGANger, I limited longer sequences to 28 readings. Neither model works with variable length sequences, meaning that some data was inevitably lost. Additionally, my experimentation with both models was limited by the length of training time, which I often ran overnight. GPU access would allow me to iterate faster, preprocess more data, train for more epochs, and perform hyperparameter optimization.

Conclusion

Despite these limitations, this study sufficiently addressed the original objectives and research questions. I answered four questions: (1) which populations are underrepresented in fNIRS research? (2) why are these populations underrepresented? (3) how does bias from underrepresentation propagate through synthetic data generation to downstream applications?, and (4) how can we improve diverse participant recruitment? Based on interviews with both researchers and participants, there are four underrepresented demographics to consider in recruitment: hair type, religious background, cultural background and understanding, and (dis)ability. Recruiting participants from these demographics can be challenging for a number of reasons, including self-exclusion, inability of participants to afford time off from work, lack of standard best practices in a developing field, and increased costs of working with curly and thick hair. The synthetic data evaluation showed that underrepresentation in training data propagates in the generated data, decreasing both the data quality and data utility. More inclusive recruitment methods will help increase the diversity of the training set, thereby increasing the quality of synthetic data – this will help to decrease the bias that is propagated through machine learning methods, improve medical trust, and improve existing best practices.

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Appendix A – Interview Notes

Demographics

- Education: two have a B.S., three are in-progress PhD students, and two have completed their PhD
- Race: Two east Asian, one black, one south Asian, and three white
- Nationality: five U.S. citizens, two non-U.S. citizens
- Gender: one female, and six male
- Hair Type: one curly hair, one bald, one long hair
- Exposure to fNIRS: five researchers, two participants

Colby

- Recent graduate of CU Boulder's B.S. Computer Science program
- Interned on an AI team building computer vision models for 2.5 years; did not work directly on fairness, but was mostly concerned with data gathering to ensure that proprietary sources were not used
- Also worked to ensure there were no data breaches
- Reported that he would do the fNIRS research if he felt like it was really important
- Thought cornrowing was a good idea – shows the researchers had put in the effort to learn about black hair and how it works, and that they had the forethought to use a professional
    - Has cornrowed his hair before – would do it for a research project, but not for daily wear
- Question: Should participants be paid more if their research process involves additional steps, such as cornrowing the hair?
    - Answer: Could see an argument for it, but also, it could be used as a reason to skip over curly haired people
    - Suggested that coming in with it already cornrowed, or setting up a separate meeting may also be options
    - Stated that gel is realistically an easier solution

Don

- Professor at Colorado State University in the Department of Psychology
- Presented "Skin and Hair Influences on fNIRS Data"
- fNIRS literature systemically fails to report sample demographics based on skin and hair phenotype
    - Kwasa 2023
    - Pringle 1999 suggests that hair color under optodes is a big predictor of quality
    - Hair thickness/length/curl/melanin are all relevant factors
- His study's aim is to show impact on tasks – sequential finger opposition and n-back
- Regression where x if fNIRS variable and y is hair or skin predictor
- Factors that are correlated: hair curliness, hair melanin, location of head (prefrontal cortex is better than the top of the head, for example)
- Factors not correlated: Monk skin tone measure, melanin index, individual typology analogy, ITA hair color

Jason

- Recent graduate of CU Boulder's B.S. Computer Science program
- Has participated in fNIRS research previously
- Would be more likely to participate if he knew the experiment involved wearing a cap (enjoyed wearing the cap in the last research session)
- No fairness concerns
- No personal challenges with the experiment
- The setup seemed to take the longest, with the preparation and then getting it to sit correctly on the head

Jeremy

- Scientific consultant at NIRx, the largest manufacturer of fNIRS equipment
- Presented "fNIRS Workshop Introduction"
- Best Practices for fNIRS Publications: part of efforts to standardize techniques/analysis
- Asked about datasets where individual demographics are labelled. Responded that there may be one or two, but that it certainly is not common practice
- Additional Notes: Recently co-authored "Interdisciplinary Views of fNIRS: Current Advancements, Equity Challenges, and an Agenda for Future Needs of a Diverse fNIRS Research Community", but he did not bring this up in any of our conversations? I found this on his Google Scholar profile when I was looking at his background for the demographics.

Leo

- PhD student in CU Boulder's joint Computer Science/ATLAS/Neuroscience programs
- fNIRS research in mixed reality
- Referred me to his recent paper about driving in VR
- Performed previous simulations with audio, tactile, and visual cues
- Not intended for fairness/accessibility purposes, but could think of applications (such as using tactile stimulation instead of asking the participant to look at something)

Mohsena

- Working on Professor Kim's chatgpt study
    - Focuses on developing a story with parents and children, mainly on that. Kids were participating in the fNIRS data collection, more concerned if kids were comfortable or not than anything else
    - No fairness considerations
    - Had a person dedicated to recruitment. IRB approved. Human pool of participants. Not sure 100% how recruitment is done, but does know there's a pool they call from
    - Developing protocol for the study. After it was done, would read the child's story, then generate stories for them
    - Major challenge: we understand technical background and virtual characters. But the participants and kids/parents did not always understand. Concerned about the privacy of the study - kids don't understand what to say, so they might say sensitive

stuff about the family. Redirect ChatGPT to follow the instructions - don't say bad words, use simple language, etc. Don't say the history, etc.

- Working on scalable LLM agents now
- Muslim women
    - Previously, we worked on inclusive fNIRS together in an attempt to make it accessible for Muslim women [15]
    - Tom (our advisor) thought that it would be hard to gather participants - could be issues with Muslim countries/people, so decided to leave it as it is (after the pilot)
    - Broad aspect of study if it can be done - people are not familiar with fNIRS itself. large part of population is lacking from these studies
    - Muslim women who are open minded could be interested in the study, but to proceed, would need to make sure their privacy is intact - separate environment, etc. Other than that, if someone is reserved/conservative, may not be interested
    - Professor Kim is trying to run another study - one teenager was interacting with AI agent, and became so isolated/lonely that he committed suicide. Most LLM agents don't announce themselves as such. Motivated her to study chatbot conversations with human participants, and some qual/quant analysis on that mode of conversation
- AI agents should be transparent

Nolan

- PhD student in CU Boulder's Computer Science program
- fNIRS research primarily focused on pregnant people and infants
- Not explicit consideration of fairness or representation in fNIRS
- Not done too much ML stuff in fNIRS or neuroimaging in general
- RISE study: tried to balance the sample
    - Recruiting underrepresented populations such as racial diversity, socioeconomic background, etc.
    - Not to train unbiased ML model
    - To make sure capturing correct neural feedback/underpinnings
    - All women, sampling from Denver
    - 60% white, 20% black, long-tail distribution of others (Native american, pacific islander, etc.)
    - Based purely on time it takes to do studies, usually people who are more comfortable. Single mother of three is probably not coming in to do this, it's more dual-income demographics
- fNIRS studies, it's a finite amount of time and IRB makes it difficult to get people outside certain procedures. left sampling classes or putting up flyers, so control is relatively small. neuroimaging you never really get a representative dataset, unless you're willing to sacrifice power for that. big studies are 100-150 people only
- Channels: depends on criteria. for MNE, looking at scalp coupling index. Signal to noise ratio function above 0.5, throw it out (based on literature). As far as data, it's really bad - would need to throw out almost every channel for 60% of the participants because they're

kids (nothing to do with hair type). movement in kids makes it difficult to sit still and collect good data

- A few techniques to work pretty well - algorithmic filtering techniques to get rid of the need to remove channels. Not in an effort for fairness, but for completeness and data preservation
- As far as our scans, we have to turn people away because they had too much hair. really hard to scan people with thick, curly hair
- for mri it's less sensitive
- think that cornrowing is great, but we're one of two labs at CU that uses fNIRS. there's 0 chance we get hairstylist to come in for our sessions not because it's problematic but it's expensive
- Nolan is curious about fNIRS's role in neuroimaging research
    - Could be really convenient way of doing proof of concept research (proposal for big MRI study, but pilot on fNIRS to prove there's some sort of result). Exploratory stuff for cheaper
    - In that sense, a lot of studies would be unfunded, but could be even more problematic
    - EEG does temporal resolution better than fNIRS, and MRI does spatial resolution better than fNIRS
    - fNIRS does good spatial resolution for really low physiological response, so we can't use it for BCI. fNIRS is like a souped-up mini cortical MRI
    - No granularity and no reference point
    - Would not want to do fNIRS research point blank, but thinks it's a great way to collect a lot of data efficiently and cheaply. Which is why it will always be unfunded
- Lots of algorithmic work in fNIRS/EEG/MRI is in an effort to reduce noise, but fairness is a secondary benefit. PCA is becoming relatively popular for filtering noise
- Problematic: use cases are potentially problematic, such as drawing inferences from data and generalizing
    - More likely to find results on bald, white people. Get high rates of false negatives in curly hair and darker skin, since you're likely to throw out more signal/noise, so it inadvertently holds back findings
    - Noisier data -> more likely to exclude signal perceived as noise